Report 2

Authors
Tesnim Omran, Diana Dawidi, Adrian Nasrat, Ibrahim Mizher

Group number
4

Date
27/12/25

Report 2

1 Description of business

Nordic Learning Solutions (NLS) is a digital learning platform aimed at supporting educational institutions through personalized and data-driven learning experiences. The platform is designed for students, teachers and administrators, the system provides functionality for user registration, course management, learning progress tracking and AI-based learning recommendations.

The main objective of the system is to improve learning quality and efficiency by automating administrative tasks, centralizing learning content & student data and analyzing learning behavior to generate personalized recommendations. By using structured data and analytics, NLS supports both students in their learning journey and teachers in managing courses and content more effectively.

2 Concept model and comparison with the ER

2.1 Concept model

In this project, Sowa's conceptual model is used as a theoretical foundation for understanding and structuring the application domain. The model guides the identification of key concepts and relationships, such as Student, Course, Learning Event, and Recommendation, which are represented concretely in the Entity–Relationship (ER) model.

Sowa's approach emphasizes semantic meaning and real-world interpretation, allowing concepts to be defined independently of technical implementation. This makes it suitable for analyzing the learning environment at a conceptual level and supporting communication between stakeholders, while the ER model provides a more formal and implementation-oriented representation of the same concepts.

We chose Sowa because it focuses on semantic meaning and real-world interpretation, which fits well with understanding our learning platform before technical implementation.

2.2 Comparison with the ER Model

The ER model represents the same domain using entities, attributes and relationships, focusing on data structure and consistency. While the ER model is precise and suitable for database design, it is more technical and less expressive in capturing semantic meaning. The Entity–Relationship model used to describe the data structure is shown in Appendix C.

In contrast, Sowa's conceptual model provides a higher-level abstraction, focusing on understanding the domain rather than implementing it. The ER model answers how data is structured for implementation, while the conceptual model focuses on meaning and interpretation.

2.3 Discussion

For this project, the conceptual model is more suitable for early analysis and communication, as it helps clarify domain understanding without technical constraints. The ER model becomes more useful in later stages when the system is implemented and data structures must be precisely defined.

An improvement would be to strengthen traceability between conceptual concepts and ER entities, ensuring that all important domain concepts are consistently represented in the data model.

3 Process model and comparison with DFD

3.1 Process model

In addition to the Data Flow Diagram, a workflow-based process is used to describe the learning process. The workflow illustrates the sequence of activities from user registration and course enrollment to learning activity, progress tracking and generation of personalized recommendations.

The workflow includes validation steps to handle unsuccessful actions, making the learning process iterative and realistic. The workflow-based process model is illustrated in Appendix E.

3.2 Comparison with DFD

The DFD focuses on how data moves between processes, data stores and external actors, while the workflow focuses on the order and execution of activities over time. The Data Flow Diagram describing the system processes is presented in Appendix B.

The DFD is effective for understanding information flow and system boundaries, whereas the workflow provides a clearer view of process logic and dependencies between activities.

3.3 Discussion

Using both models provides complementary perspectives. The DFD supports functional analysis and data management, while the workflow improves understanding of operational behavior.

An improvement would be to align workflow steps more explicitly with DFD processes to ensure consistency across models.

4 Pros and cons of OM and EM

4.1 Pros

The Objective Model provides a clear structure for identifying goals, obstacles, causes and solutions, which helps justify system requirements. The Objective Model is shown in Appendix A.

The Enterprise Model integrates objectives, actors, processes, data and applications, enabling traceability across different modelling perspectives. The Enterprise Model linking objectives, processes, data, and applications is shown in Appendix D.

This traceability also supports consistency across modelling perspectives.

4.2 Cons

The Objective Model may oversimplify complex organizational dynamics by focusing on a limited set of goals and obstacles.

The Enterprise Model can become difficult to read if too many elements are unlabeled and included.

4.3 Improvements

The Objective Model could be improved by introducing measurable indicators to evaluate whether goals are being achieved in practice.

The Enterprise Model could be improved by further standardizing relationship labels and limiting the number of elements to maintain clarity while preserving traceability.

5 Conclusion

This report has presented a conceptual analysis of Nordic Learning Solutions using multiple modeling approaches introduced in the course. By applying the Objective Model (OM), Data Flow Diagram (DFD), Entity-Relationship (ER) model and Enterprise Model (EM), the business and its supporting information system were described from complementary perspectives.

The use of these models made it possible to connect organizational goals with processes, data and system requirements, ensuring traceability across different abstraction levels. While each model has its own strengths and limitations, their combined use provides a more complete and structured understanding of the enterprise.

Overall, the modeling work demonstrates how conceptual models can support both analysis and communication in information systems development, while also highlighting areas where clarity and consistency can be further improved.

6 References

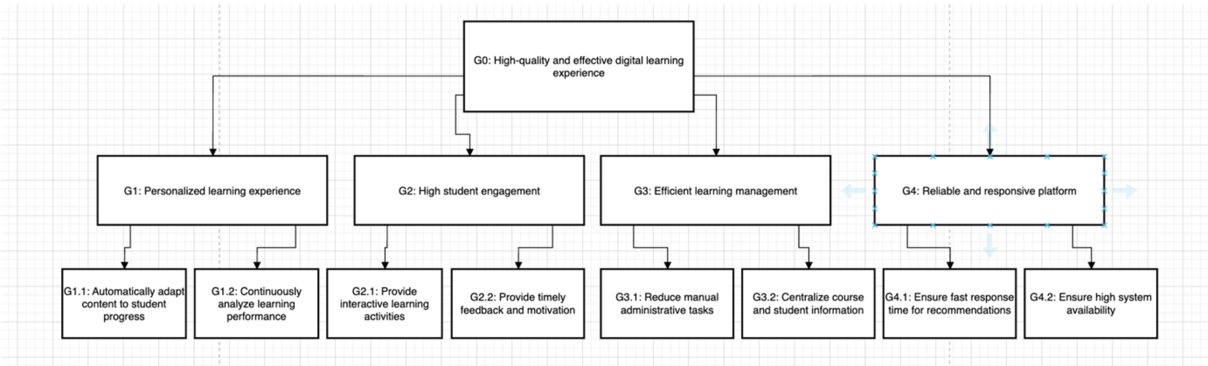
Bubenko, J. A. (1995). *Requirements modelling in information systems development*. IFIP WG8.1 Working Conference on Information Systems Development.

Langefors, B. (1995). *Essays on infology: Summing up and planning for the future*. Studentlitteratur.

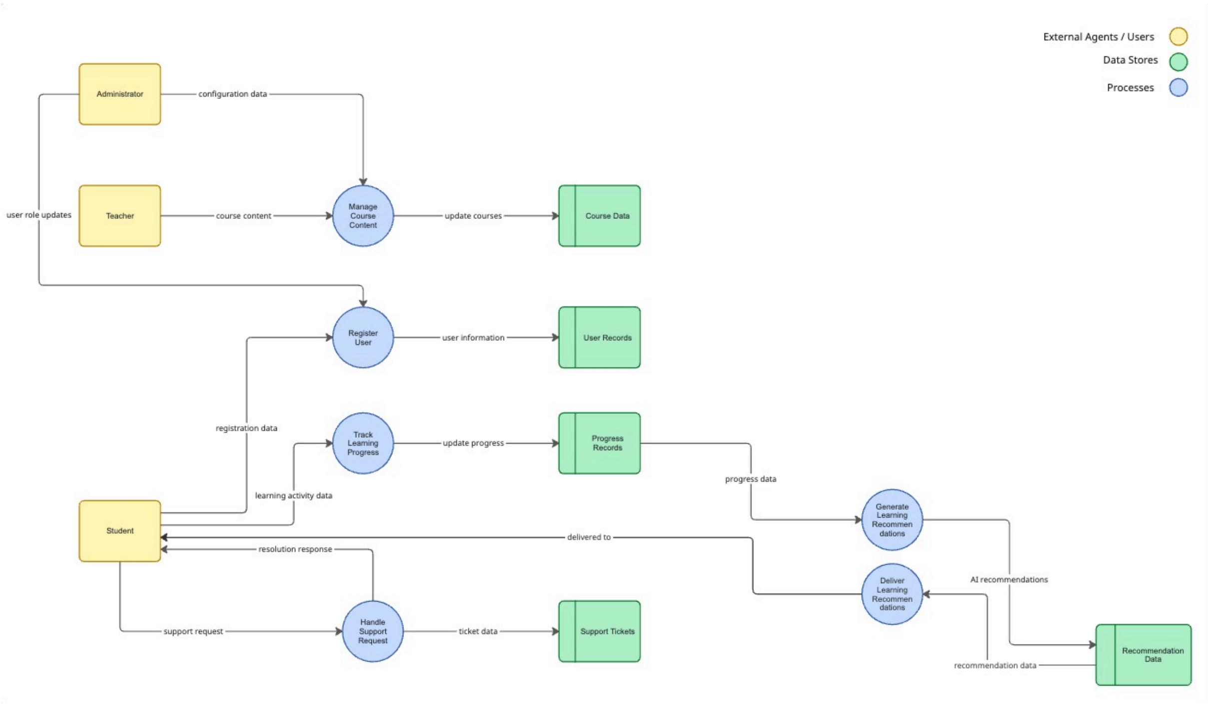
Sundgren, B., & Steneskog, G. (2005). Information systems for concerted actions. In *Exploring patterns in information management: Concepts and perspectives for understanding IT-related change*. Ekonomiska forskningsinstitutet vid Handelshögskolan i Stockholm (EFI).

Sowa, J. F. (2000). *Knowledge representation: Logical, philosophical, and computational foundations*. Brooks/Cole.

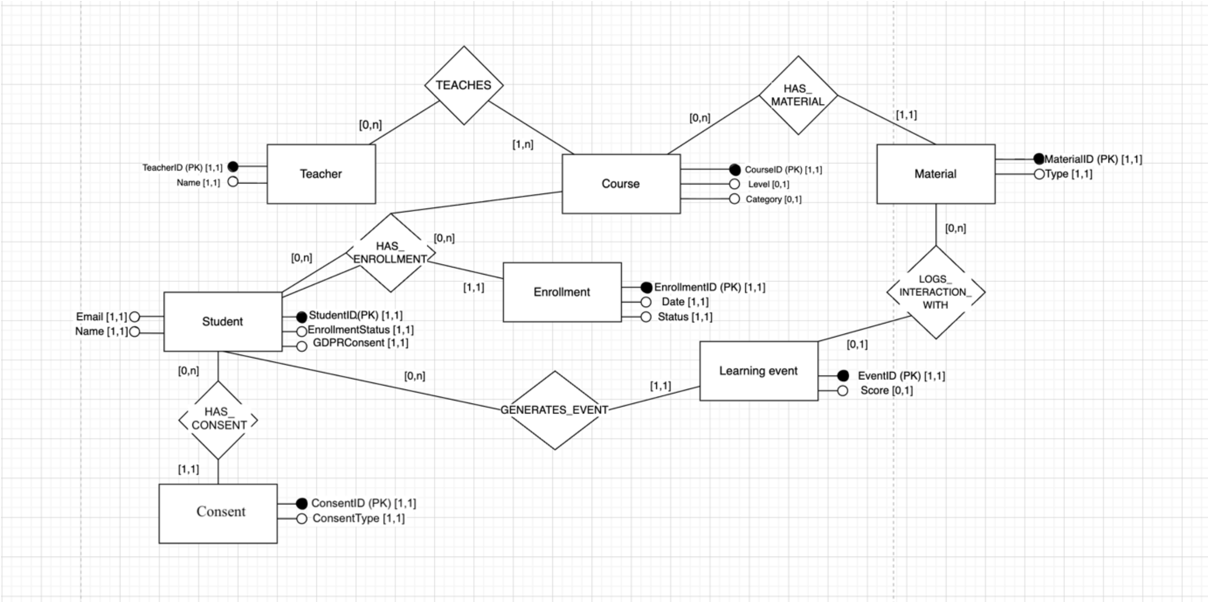
Appendix A – Objective Model (OM)



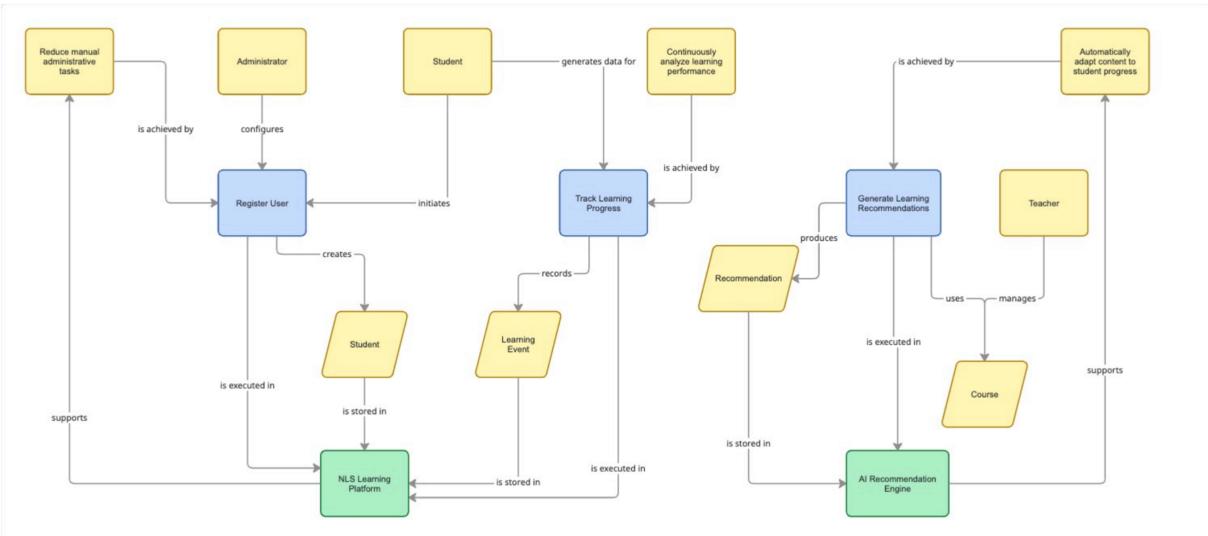
Appendix B – Data Flow Diagram (DFD)



Appendix C – Entity–Relationship Model (ER)



Appendix D – Enterprise Model (EM)



Appendix E – Workflow Model